

DEPARTMENT OF COMPUTER SCIENCE
Mobile Application Development



MYSTERY BOX APP

Project Report

Submitted By	Abdul Rehman & Muhammad Nadeem
Subject	Mobile Application Development
Teacher	Sir Faisal Hafeez

1. Introduction

The **Mystery Box App** is an Android application designed to send surprise wishes in a fun and creative way. Users can create a mystery box containing text messages, images, and videos. The receiver opens the mystery box and experiences a surprise presentation of the shared content.

The application aims to make communication more creative, memorable, and enjoyable for users during special occasions.

2. Project Objectives

The main objectives of the project are:

- To create a fun and interactive Android application.
- To allow users to send surprise wishes digitally.
- To provide support for multimedia content such as text, images, and videos.
- To make communication more creative and memorable.
- To learn Android application development with Firebase integration.

3. Scope of the Project

The Mystery Box App can be used for various purposes, providing an innovative way to share emotions and messages with friends and family:

Use Case	Description
Birthday Wishes	Send surprise animated birthday greetings
Eid Greetings	Creative Eid messages with multimedia
Friendship Messages	Fun boxes for friends on special days
Party Invitations	Surprise invitation reveals
Anniversary Surprises	Romantic multimedia boxes
Special Event Wishes	Any celebration or milestone

4. System Architecture

User → Android App → Firebase Backend → Receiver

Working Process:

Step 1	User creates a mystery box.
Step 2	User adds text, pictures, or videos.

Step 3	Content is uploaded to Firebase.
Step 4	A shareable link is generated.
Step 5	The receiver opens the mystery box.
Step 6	Surprise content is displayed.

5. Technologies Used

Category	Technology / Tool	Purpose
Frontend	Android Studio	IDE for development
Frontend	XML Layouts	UI design
Frontend	Java / Kotlin	Programming language
Backend	Firebase Realtime Database	Data storage
Backend	Firebase Storage	Media file storage
Backend	Firebase Authentication	User login (if implemented)
Tools	Android SDK	Android development kit
Tools	Firebase Console	Backend management
Tools	GitHub	Version control

6. Features of the Application

- Create mystery boxes with ease.
- Add custom text messages.
- Upload images to the mystery box.
- Upload videos to the mystery box.
- Generate shareable links for receivers.
- Surprise content display on opening.
- Easy and user-friendly interface.

7. Challenges Faced

During development, the following challenges were encountered:

- Firebase integration issues during initial setup.
- Uploading large video files efficiently.
- Device compatibility problems across Android versions.

- Internet speed limitations during media uploads.

8. Risks and Solutions

Risk	Solution
Slow internet affecting performance	Regular testing & optimization
Large media files require more storage	Image/video size compression
Firebase costs at scale	Firebase secure storage controls
App crashes on lower-end devices	Continuous bug fixing & debugging

9. Limitations

The current version of the application has some limitations:

- Requires an active internet connection.
- Limited offline functionality.
- Large videos may upload slowly on weak networks.
- Dependency on Firebase services for core features.
- Available only for Android devices currently.

10. Future Scope

The following features can be added in future versions:

- User accounts and profiles.
- In-app chat system.
- Voice messages support.
- Cloud storage integration.
- Push notifications for box received alerts.
- Dark mode interface.
- AI-generated surprise messages.

11. Conclusion

The **Mystery Box App** successfully provides a unique and creative platform for sending surprise wishes. The application supports multimedia content and uses Firebase for reliable backend services. Through this project, valuable knowledge of Android development, Firebase integration, and mobile application design was gained.

Future enhancements can further improve user experience, extend platform support to iOS, and add AI-driven features to make the app even more engaging.

. References

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